

MENGER DIAGONAL SLICES: A RECURRENCE OF ORDER $\lceil D/2 \rceil$

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ABSTRACT. Slice a Menger sponge along its main diagonal and count the cubes the cut meets; then do the same in every dimension. The count obeys a linear recurrence with constant coefficients of order at most $\lceil D/2 \rceil$ in dimension D , a fourfold sharpening of the order $2D + 1$ that the standard carry automaton gives for free; exactness of that order is verified for $2 \leq D \leq 24$. At $D = 3$ the machine outputs $\begin{pmatrix} 6 & 6 \\ 1 & 3 \end{pmatrix}$, reproducing the published hexagon-triangle substitution spectrum and the integers 1, 6, 42, 306, 2250, 16578, 122202 by a route that never mentions a hexagon.

1. INTRODUCTION

Take a cube. Cut it into 27 equal subcubes and throw away the seven that touch the middle of a face or the middle of the cube; twenty survive. Repeat inside each survivor forever, and the limit is the Menger sponge. Now slice the sponge with the plane through its centre perpendicular to a main diagonal. The picture that falls out is famous: a hexagon full of six-pointed stars, first rendered by Pérez-Duarte [9], popularised by Hart [4] and by Cook's slicing code [3], and given its substitution rule and its dimension 1.8184 by Abel [1].

This paper asks the same question one dimension at a time. In dimension D the sponge becomes the set of cells with at most one coordinate digit equal to the middle digit 1, and the diagonal cut becomes the hyperplane $\sum_i x_i = D(3^L - 1)/2$ through the centre of the level- L grid. Let $a_D(L)$ count the cells the hyperplane meets. The question is how much memory this sequence has: how many previous terms determine the next one.

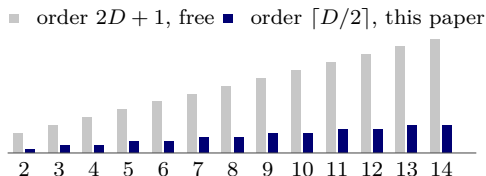


FIGURE 1. How much memory the slice census needs, by dimension D .

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The answer is Theorem 3.1: at most $\lceil D/2 \rceil$ terms, in every dimension at once. Figure 1 is the whole contribution in one picture. Three moves do it. First, the digit polynomial factors,

$$P_D(t) = (1 + t^2)^{D-1}(1 + Dt + t^2),$$

which is a two-line count of how a kept digit vector can sum to s . Second, the carry map $c \mapsto (c + D - s)/3$ contracts: its fixed point is $D/2$, integrality is a wall, and every carry reachable from zero obeys $|c| \leq \lfloor (D-1)/2 \rfloor$. Third, the design is invariant under $v \mapsto (2, \dots, 2) - v$, so $P_D[s] = P_D[2D - s]$, the carry reflection $c \mapsto -c$ commutes with the transfer matrix, and the orbit of the start state lives inside its $+1$ eigenspace, whose dimension is $\lfloor (D-1)/2 \rfloor + 1 = \lceil D/2 \rceil$.

What is not new. That some constant-coefficient recurrence exists is free and standard. A finite carry state space makes $a_D(L)$ a 3-regular sequence in the sense of Allouche and Shallit [2], and the untrimmed automaton on states $-D, \dots, D$ already hands over order $2D + 1$ with no argument at all. A paper that led with “these slices obey a linear recurrence” would be selling a textbook fact. The object is not new either: higher-dimensional Menger continua are classical, and Hocking [5] cuts exactly the $D = 4$ diagonal for its geometry and its pictures. The $D = 3$ integers are catalogued as OEIS A299916 [8], whose Menger reading is a user comment counting the star-shaped holes rather than the tiles.

What is new. The exact factorisation of P_D , the sharp order $\lceil D/2 \rceil$ proved for every $D \geq 2$, the exactness of that order verified through $D = 24$, the closed form of the reduced matrix, and every census at $D \geq 4$. Roughly a factor of four, uniformly in the dimension.

The cold anchor. At $D = 3$ the construction, which never mentions a hexagon or a triangle, returns the 2×2 integer matrix $\begin{pmatrix} 6 & 6 \\ 1 & 3 \end{pmatrix}$, characteristic polynomial $\lambda^2 - 9\lambda + 12$, dominant root $(9 + \sqrt{33})/2$ and census 1, 6, 42, 306, 2250, 16578, 122202. Abel’s tile substitution $\begin{pmatrix} 6 & 1 \\ 6 & 3 \end{pmatrix}$ [1] has the same trace 9 and determinant 12, A299916 has signature $(9, -12)$ and the same integers, and $\log_3((9 + \sqrt{33})/2) = 1.818410$ is Abel’s published dimension. Two different routes to one spectrum is the reason to believe the machine in the dimensions where nobody has looked.

2. DEFINITIONS

Definition 2.1. The D -dimensional Menger design is

$$F_D = \{v \in \{0, 1, 2\}^D : \#\{i : v_i = 1\} \leq 1\},$$

the digit vectors with at most one middle coordinate. Its *fill* is $|F_D|$.

Definition 2.2. For $L \geq 0$ write each $x_i \in \{0, \dots, 3^L - 1\}$ in base 3 as $x_i = \sum_{j=0}^{L-1} v_i^{(j)} 3^j$. The *level- L Menger analog* is

$$S_L^{(D)} = \{x : v^{(j)} \in F_D \text{ for every } j\},$$

and the *central diagonal census* is

$$a_D(L) = \#\{x \in S_L^{(D)} : \sum_i x_i = D(3^L - 1)/2\}, \quad a_D(0) = 1.$$

The hyperplane is the one through the centre of the grid: the coordinate sum of the centre cell of $\{0, \dots, 3^L - 1\}^D$ is $D(3^L - 1)/2$. At $D = 3$ this is the diagonal cut of the classical sponge.

Definition 2.3. The *digit polynomial* is

$$P_D(t) = \sum_{v \in F_D} t^{v_1 + \dots + v_D} = \sum_{s=0}^{2D} B_D(s) t^s.$$

Definition 2.4. The *transfer matrix* $M^{(D)}$ acts on carry states $c \in \mathbb{Z}$ by

$$M_{c'c}^{(D)} = B_D(c + D - 3c'),$$

which is nonzero only when $0 \leq c + D - 3c' \leq 2D$. The *core* is $C_D = \{c \in \mathbb{Z} : |c| \leq r_D\}$ with $r_D = \lfloor (D-1)/2 \rfloor$, and $M_{\text{even}}^{(D)}$ is the matrix of $M^{(D)}$ restricted to the core and expressed in the reflection-symmetric basis $v_0 = e_0$, $v_k = e_k + e_{-k}$ for $1 \leq k \leq r_D$. Its size is $n_D = r_D + 1 = \lceil D/2 \rceil$, and χ_D denotes its characteristic polynomial.

3. RESULTS

Theorem 3.1. For every $D \geq 2$ the sequence $a_D(L)$ satisfies the linear recurrence with constant integer coefficients given by χ_D , the characteristic polynomial of $M_{\text{even}}^{(D)}$. In particular a_D satisfies a constant-coefficient linear recurrence of order at most $\lceil D/2 \rceil$.

Lemma 3.2. For every $D \geq 1$, $P_D(t) = (1+t^2)^{D-1}(1+Dt+t^2)$, the fill is $|F_D| = 2^{D-1}(D+2)$, and

$$B_D(2k) = \binom{D}{k}, \quad B_D(2k+1) = D \binom{D-1}{k},$$

with the convention that a binomial coefficient with an out-of-range lower index is zero.

Lemma 3.3. Every carry state reachable from $c = 0$ lies in the core C_D , and the core is forward invariant: if $|c| \leq r_D$ and $M_{c'c}^{(D)} \neq 0$ then $|c'| \leq r_D$.

Lemma 3.4. $P_D[s] = P_D[2D-s]$ for all s , the reflection $R : e_c \mapsto e_{-c}$ satisfies $RM^{(D)} = M^{(D)}R$, and $Re_0 = e_0$. Consequently the orbit $\{(M^{(D)})^L e_0 : L \geq 0\}$ lies in the $+1$ eigenspace of R inside the core, a space of dimension $\lceil D/2 \rceil$.

Lemma 3.5. Let $\omega = e^{2\pi i/3}$. Then $P_D(\omega) = (-1)^{D-1}\omega^D(D-1)$, so $|P_D(\omega)| = D-1$ for every $D \geq 1$.

Lemma 3.6. Write $\varepsilon_D = (D - 1)(-1)^{D-1}/3$ and $f_D = 2^{D-1}(D + 2)$. For $j \in \{0, 1, 2\}$,

$$\sum_{s \equiv j \pmod{3}} B_D(s) = \begin{cases} f_D/3 + 2\varepsilon_D, & j \equiv D \pmod{3}, \\ f_D/3 - \varepsilon_D, & \text{otherwise.} \end{cases}$$

Proposition 3.7. Fix an integer offset m with $|m| \leq r_D$ and let $a_D(L; m)$ count the cells of $S_L^{(D)}$ on the parallel hyperplane $\sum_i x_i = D(3^L - 1)/2 + m$. Then $a_D(\cdot; m)$ satisfies the same recurrence as a_D , the one given by χ_D , and $a_D(L; m) = a_D(L; -m)$ for every L .

Remark 3.8. The hypothesis $|m| \leq r_D$ is sharp at even D . At $D = 4$ the offset $m = 2 = D/2$ sits exactly on the fixed point of the carry map, never enters the core, and the residual of χ_4 applied to $a_4(\cdot; 2)$ is the constant 128 rather than zero, for every window checked by `scripts/verify.py`.

Corollary 3.9. $\limsup_L a_D(L)^{1/L}$ is the modulus of a root of χ_D , hence an algebraic number of degree at most $\lceil D/2 \rceil$. Whenever the counting exponent $\dim_{\text{slice}}(D) = \lim_L \log_3 a_D(L)/L$ exists, it is $\log_3$ of that algebraic number.

Fact 3.10. For every D with $2 \leq D \leq 24$ the Hankel determinant $H_D = \det[a_D(i+j+1)]_{0 \leq i, j < n_D}$, computed in exact rational arithmetic, is nonzero; hence the minimal order of a_D is exactly $\lceil D/2 \rceil$ for those D . The first values are $H_2 = 2$, $H_3 = 72$, $H_4 = -6336$, $H_5 = -1029600000$, $H_6 = -62272025640000$, and H_{24} is a nonzero integer of 899 digits. Checked by `scripts/verify.py`.

Fact 3.11. For every D with $2 \leq D \leq 24$ the set of carry states reachable from $c = 0$ is exactly the core $\{c : |c| \leq \lfloor (D-1)/2 \rfloor\}$, of size $D-1$ at even D and D at odd D . Checked by `scripts/verify.py`.

Fact 3.12. For every D with $2 \leq D \leq 24$,

$$\text{tr } M_{\text{even}}^{(D)} = \begin{cases} 3 \cdot 2^{D-2} - 1, & D \text{ even,} \\ 3D \cdot 2^{D-3}, & D \text{ odd.} \end{cases}$$

The even values are 2, 11, 47, 191, 767, 3071, 12287, ... and the odd values are 9, 60, 336, 1728, 8448, 39936, Checked by `scripts/verify.py`.

Fact 3.13. At $D = 3$, $M_{\text{even}}^{(3)} = \begin{pmatrix} 6 & 6 \\ 1 & 3 \end{pmatrix}$ with $\chi_3(\lambda) = \lambda^2 - 9\lambda + 12$, dominant root $(9 + \sqrt{33})/2$ and census 1, 6, 42, 306, 2250, 16578, 122202, matching OEIS [A299916](#) term for term and its signature $(9, -12)$, and $\log_3((9 + \sqrt{33})/2) = 1.818410$ to six places. At $D = 2$ the census is 2^L , of order 1. At $D = 4$, $\chi_4(\lambda) = \lambda^2 - 11\lambda - 66$ with dominant root $(11 + \sqrt{385})/2$ and census 1, 6, 132, 1848, 29040, 441408, 6772128. At $D = 5$ and $D = 6$ the censuses begin 1, 30, 1000, 35700, 1321600 and 1, 20, 4030, 242300, 24642700. Checked by `scripts/verify.py`, which also confirms that three independent generators, brute-force digit enumeration, the substitution product $\prod_{j < L} P_D(t^{3^j})$,

and the untrimmed carry automaton on states $-D, \dots, D$, agree term for term for $2 \leq D \leq 7$ and $0 \leq L \leq 4$.

Conjecture 3.14. *For every $D \geq 2$ the counting exponent sits on alternating sides of the solid's own dimension minus one, $\log_3 f_D - 1 = \log_3(f_D/3)$:*

$$\operatorname{sgn}(\rho_D - f_D/3) = (-1)^{D+1}, \quad f_D = 2^{D-1}(D+2),$$

where ρ_D is the dominant root of χ_D . The comparison value $\log_3 f_D - 1$ is the generic one: for a set of Hausdorff dimension s in $\mathbb{R}^D$, almost every hyperplane meets it in dimension at most $s - 1$, with equality on a positive-measure set of translates once $s > 1$ [6, 7], and $\log_3 f_D$ is the dimension of the solid. The central diagonal hyperplane is one specific, highly symmetric slice, to which no such theorem applies; the conjecture is that its counting exponent misses the generic value in every dimension, alternately above and below. Evidence: the sign alternates without exception for $2 \leq D \leq 20$ computed from the roots of χ_D , and the exact rational sign of $\chi_D(f_D/3)$ alternates as $(-1)^D$ for $2 \leq D \leq 30$, both in `scripts/verify.py`; the two agree wherever they overlap, and the second uses no floating point at all. Failure mode: the exact determinant sign is the sign of $\prod_i (f_D/3 - \lambda_i)$ over the whole spectrum, so it reports on ρ_D alone only while every other eigenvalue satisfies $|\lambda_i| < f_D/3$. That side condition is itself unproved, and Conjecture 3.15 says any proof of it must be delicate. A single dimension where a subdominant eigenvalue crosses $f_D/3$ would break the bridge between the two pieces of evidence without disturbing either computation.

Conjecture 3.15. *There is no uniform spectral gap: writing λ_2 for a second-largest eigenvalue of $M_{\text{even}}^{(D)}$ in modulus,*

$$\frac{\rho_D}{|\lambda_2|} = \frac{D+2}{D-2} + O(D^{-3}) \longrightarrow 1.$$

Evidence: the computed ratios fall from 3.551785 at $D = 4$ to 1.222222 at $D = 20$, monotonically from $D = 6$ on; over $6 \leq D \leq 20$ they agree with $(D+2)/(D-2)$ to within 0.05, and to six decimal places by $D = 18$. The two smallest dimensions in that range sit outside the agreement, 3.551785 against 3 at $D = 4$ and 2.253050 against 2.333333 at $D = 5$, which is where the $O(D^{-3})$ correction is largest. `scripts/verify.py` asserts the monotonicity, and the agreement over $6 \leq D \leq 20$. Failure mode: the ratios are floating-point roots of exactly computed integer polynomials, so a genuine near-degeneracy at some larger D could be misread; and the fitted shape $(D+2)/(D-2)$ is an observed match, not a derivation. This is the lighthouse of the paper. Any argument that pins the subdominant spectrum below $f_D/3$, which is what Conjecture 3.14 needs, has to work inside a margin that shrinks like $4/D$, and so cannot be a fixed-size perturbation estimate. Four natural routes were tried and refuted by their own computations: $M_{\text{even}}^{(D)}$ is not banded plus low rank, not totally nonnegative, not symmetrizable by a positive diagonal, and $f_D/3 \cdot I - M_{\text{even}}^{(D)}$ is not an M -matrix.

4. PROOFS

Proof of Lemma 3.2. Split F_D by the number of middle coordinates. A vector with no coordinate equal to 1 chooses 0 or 2 in each of the D places, contributing $(1+t^2)^D$ to the generating function. A vector with exactly one coordinate equal to 1 chooses which coordinate, in D ways, and then 0 or 2 in the remaining $D-1$ places, contributing $Dt(1+t^2)^{D-1}$. No other vector lies in F_D , so

$$P_D(t) = (1+t^2)^D + Dt(1+t^2)^{D-1} = (1+t^2)^{D-1}(1+Dt+t^2).$$

Setting $t = 1$ gives $|F_D| = 2^{D-1}(D+2)$. For the coefficients, expand $(1+t^2)^{D-1}(1+t^2) = (1+t^2)^D$ for the even part and $Dt(1+t^2)^{D-1}$ for the odd part: the coefficient of t^{2k} is $\binom{D}{k}$ and that of t^{2k+1} is $D\binom{D-1}{k}$. $\square$

Proof of Theorem 3.1. Step 1: the census is a coefficient. By Definition 2.2 a cell of $S_L^{(D)}$ is exactly a choice of $v^{(0)}, \dots, v^{(L-1)} \in F_D$, and its coordinate sum is $\sum_i x_i = \sum_{j < L} 3^j \sigma_j$ with $\sigma_j = \sum_i v_i^{(j)}$. The number of vectors in F_D with digit sum σ is $B_D(\sigma)$, so for any target T ,

$$\#\{x \in S_L^{(D)} : \sum_i x_i = T\} = [t^T] \prod_{j=0}^{L-1} P_D(t^{3^j}).$$

Step 2: the carry recursion. For $c \in \mathbb{Z}$ let $N_L(c) = [t^{T_L(c)}] \prod_{j < L} P_D(t^{3^j})$ with $T_L(c) = D(3^L - 1)/2 + c$, so $a_D(L) = N_L(0)$ and $N_0(c) = \mathbf{1}[c = 0]$. Peel the least significant digit vector, of digit sum s . Since

$$D(3^L - 1)/2 = 3 \cdot D(3^{L-1} - 1)/2 + D,$$

the remaining target after dividing by 3 is $T_{L-1}(c')$ with $c' = (c + D - s)/3$, and the peel is possible exactly when $3 \mid c + D - s$. Therefore

$$N_L(c) = \sum_{\substack{0 \leq s \leq 2D \\ s \equiv c+D \pmod{3}}} B_D(s) N_{L-1}\left(\frac{c + D - s}{3}\right),$$

which in the notation of Definition 2.4 reads $N_L = (M^{(D)})^L$ applied in the sense $a_D(L) = ((M^{(D)})^L)_{00}$: the entry $M_{c'c}^{(D)} = B_D(c + D - 3c')$ is the weight of the step from source c to destination c' , and only finitely many entries are nonzero in each row and column because $0 \leq c + D - 3c' \leq 2D$.

Step 3: contraction. This is Lemma 3.3, proved below. It gives an invariant finite state set, the core C_D , containing the start state 0, so $a_D(L) = (M_C^L)_{00}$ where M_C is the finite matrix $M^{(D)}$ restricted to C_D .

Step 4: halving. This is Lemma 3.4, proved below. The vector $u_L = M_C^L e_0$ lies, for every L , in the $+1$ eigenspace E of the reflection R inside $\mathbb{R}^{C_D}$, and E is M_C -invariant because R and M_C commute. A basis of E is $v_0 = e_0$ and $v_k = e_k + e_{-k}$ for $1 \leq k \leq r_D$, so $\dim E = r_D + 1 = \lceil D/2 \rceil = n_D$. Let $M_{\text{even}}^{(D)}$ be the matrix of $M_C|_E$ in that basis and χ_D its characteristic polynomial,

of degree n_D . By Cayley–Hamilton, $\chi_D(M_C|_E) = 0$, hence for every $L \geq 0$

$$\sum_{k=0}^{n_D} \chi_{D,k} u_{L+n_D-k} = \chi_D(M_C|_E) u_L = 0,$$

where $\chi_D(\lambda) = \sum_k \chi_{D,k} \lambda^{n_D-k}$ with $\chi_{D,0} = 1$. Applying the linear functional $w \mapsto w_0$ to this vector identity gives

$$\sum_{k=0}^{n_D} \chi_{D,k} a_D(L + n_D - k) = 0,$$

a constant-coefficient linear recurrence of order $n_D = \lceil D/2 \rceil$ satisfied by a_D . The coefficients are integers because $M_{\text{even}}^{(D)}$ has integer entries. $\square$

Proof of Lemma 3.3. If $M_{c'_c}^{(D)} \neq 0$ then $c' = (c + D - s)/3$ for some $0 \leq s \leq 2D$, so $c + D - s$ lies in $[c - D, c + D]$ and

$$|c'| \leq \frac{|c| + D}{3}.$$

Suppose $|c| \leq r_D = \lfloor (D-1)/2 \rfloor \leq (D-1)/2$. Then

$$|c'| \leq \frac{(D-1)/2 + D}{3} = \frac{3D-1}{6} = \frac{D}{2} - \frac{1}{6} < \frac{D}{2}.$$

Since c' is an integer, $|c'| < D/2$ forces $|c'| \leq \lfloor (D-1)/2 \rfloor = r_D$, whether D is even or odd: at even D an integer below $D/2$ is at most $D/2 - 1 = r_D$, and at odd D it is at most $(D-1)/2 = r_D$. So the core is forward invariant. The start state $c = 0$ lies in the core because $r_D \geq 0$ for $D \geq 2$, and induction on the number of steps finishes the claim. $\square$

Remark 4.1. The reverse containment, that every core state is actually reached from $c = 0$, is not needed for Theorem 3.1 and is not proved here. It is Fact 3.11: verified for $2 \leq D \leq 24$ by breadth-first search, and open beyond. Only the upper bound of Lemma 3.3 carries theorem dress.

Proof of Lemma 3.4. The map $v \mapsto (2, \dots, 2) - v$ is an involution of $\{0, 1, 2\}^D$ that preserves the number of coordinates equal to 1, so it permutes F_D , and it sends digit sum s to $2D - s$. Hence $B_D(s) = B_D(2D - s)$ for all s . Now

$$\begin{aligned} M_{-c', -c}^{(D)} &= B_D(-c + D + 3c') = B_D(2D - (c + D - 3c')) \\ &= B_D(c + D - 3c') = M_{c'_c}^{(D)}, \end{aligned}$$

which is exactly $RM^{(D)}R = M^{(D)}$, and $R^2 = I$ gives $RM^{(D)} = M^{(D)}R$. Clearly $Re_0 = e_0$. Therefore $R(M^{(D)})^L e_0 = (M^{(D)})^L Re_0 = (M^{(D)})^L e_0$ for every L , so the whole orbit lies in the $+1$ eigenspace of R . Restricted to the core, that eigenspace consists of the vectors with $w_c = w_{-c}$, spanned by e_0 and $e_k + e_{-k}$ for $1 \leq k \leq r_D$; these $r_D + 1 = \lceil D/2 \rceil$ vectors are visibly independent. $\square$

Proof of Lemma 3.5. Since $\omega^2 + \omega + 1 = 0$ we have $1 + \omega^2 = -\omega$ and $1 + D\omega + \omega^2 = (D-1)\omega$. Substituting into Lemma 3.2,

$$P_D(\omega) = (-\omega)^{D-1}(D-1)\omega = (-1)^{D-1}\omega^D(D-1),$$

and $|\omega| = 1$ gives $|P_D(\omega)| = D-1$. $\square$

Proof of Lemma 3.6. Roots of unity filter: for $j \in \{0, 1, 2\}$,

$$\sum_{s \equiv j} B_D(s) = \frac{1}{3} \sum_{m=0}^2 \omega^{-jm} P_D(\omega^m) = \frac{f_D}{3} + \frac{2}{3} \operatorname{Re}(\omega^{-j} P_D(\omega)),$$

using $P_D(1) = f_D$ and $P_D(\bar{\omega}) = \overline{P_D(\omega)}$ because P_D has real coefficients. By Lemma 3.5, $\omega^{-j} P_D(\omega) = (-1)^{D-1}(D-1)\omega^{D-j}$, whose real part is $(-1)^{D-1}(D-1)$ when $j \equiv D \pmod{3}$ and $-(-1)^{D-1}(D-1)/2$ otherwise, since $\operatorname{Re} \omega^k$ is 1 for $k \equiv 0$ and $-1/2$ otherwise. Substituting $\varepsilon_D = (D-1)(-1)^{D-1}/3$ gives $f_D/3 + 2\varepsilon_D$ in the distinguished class and $f_D/3 - \varepsilon_D$ in each of the other two. $\square$

Proof of Proposition 3.7. By Step 1 and Step 2 of the proof of Theorem 3.1 with target $T_L(m)$, the off-centre census is $a_D(L; m) = ((M^{(D)})^L)_{0m} = e_0^\top (M^{(D)})^L e_m$: the source state is the offset m and the accepted final state is 0. By Lemma 3.4, R commutes with $M^{(D)}$, is symmetric, and fixes e_0 , so

$$a_D(L; -m) = e_0^\top (M^{(D)})^L R e_m = (R e_0)^\top (M^{(D)})^L e_m = a_D(L; m).$$

Hence $a_D(L; m) = e_0^\top (M^{(D)})^L w$ with $w = \frac{1}{2}(e_m + e_{-m})$. Because $|m| \leq r_D$, the vector w lies in the $+1$ eigenspace E of R inside the core, which by Lemma 3.3 and Lemma 3.4 is invariant under M_C and has dimension n_D . So $(M_C)^L w \in E$ for all L , and Cayley–Hamilton applied to $M_C|_E$ annihilates the vector sequence exactly as in Step 4; taking the coordinate functional $w \mapsto w_0$ gives the same recurrence with the same coefficients. $\square$

Proof of Corollary 3.9. By Theorem 3.1, $a_D(L) = \sum_i p_i(L) \lambda_i^L$ where the λ_i are the roots of χ_D and the p_i are polynomials. Let Λ be the largest modulus among the roots that actually occur with a nonzero polynomial; it is positive, because every entry of $M^{(D)}$ is nonnegative and so $a_D(L) = ((M^{(D)})^L)_{00} \geq (M_{00}^{(D)})^L = B_D(D)^L > 0$. Summing that expansion, the generating function $\sum_{L \geq 0} a_D(L) z^L$ is a rational function whose poles are exactly the points $1/\lambda_i$ over the occurring nonzero roots, so its radius of convergence is $1/\Lambda$, and Cauchy–Hadamard turns that into $\limsup_L a_D(L)^{1/L} = \Lambda$, the modulus of a root of χ_D . And χ_D is a monic integer polynomial of degree $n_D = \lceil D/2 \rceil$, so every root is an algebraic number of degree at most n_D . If the limit defining $\dim_{\text{slice}}(D)$ exists, then it equals $\log_3 \Lambda$ by taking logarithms. $\square$

Remark 4.2. Exactness of the order is a Hankel condition, not an eigenvalue condition. Given the upper bound n_D of Theorem 3.1, nonvanishing of $H_D = \det[a_D(i+j+1)]_{0 \leq i, j < n_D}$ forces the minimal order to be exactly n_D ; that is Fact 3.10. Indeed, if a_D obeyed some recurrence $a_D(L+m) =$

$\sum_{k=1}^m c_k a_D(L+m-k)$ of order $m < n_D$ for every $L \geq 0$, then for every $i \geq m$ row i of that matrix would be the combination $\sum_{k=1}^m c_k \cdot (\text{row } i-k)$ of earlier rows, all n_D rows would lie in the span of the first m , and H_D would vanish. Only this direction holds: a vanishing H_D proves nothing, as the sequence $5, 2, 4, 8, 16, \dots$ shows, whose minimal order is 2 because $a(L+2) = 2a(L+1)$ for every $L \geq 0$ while no order-one recurrence fits, yet whose shifted Hankel determinant $\det\begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix}$ is zero. Distinct eigenvalues would not suffice either: writing $a_D(L) = \sum_i \lambda_i^L (e_0^\top \Pi_i e_0)$ with spectral projections Π_i , exactness also needs every weight $e_0^\top \Pi_i e_0$ to be nonzero. The two-line counterexample is $M = \text{diag}(1, 2)$ with start vector $e_0 = (1, 0)$: the eigenvalues are distinct and the matrix is non-derogatory, yet the sequence $e_0^\top M^L e_0 = 1$ has order 1, not 2. The Hankel determinant sees this where the eigenvalues do not, and it is what `scripts/verify.py` computes, in exact rational arithmetic.

5. REPRODUCIBILITY

Two scripts, plain Python 3, no dependencies, no arguments, every path relative to the lane; run them from the lane root.

`scripts/verify.py` runs in about four seconds and dies loudly on the first mismatch, naming what it got and what it wanted. It covers, in order: the factorisation of Lemma 3.2 and its closed entry form against brute-force enumeration of all 3^D digit tuples for $2 \leq D \leq 8$; agreement of three independent generators, brute enumeration, the substitution product $\prod_{j < L} P_D(t^{3^j})$, and the untrimmed carry automaton on states $-D, \dots, D$, for $2 \leq D \leq 7$ and $0 \leq L \leq 4$; the exact reachable carry set by breadth-first search for $2 \leq D \leq 24$, which is Fact 3.11; nonvanishing of the exact rational Hankel determinant for $2 \leq D \leq 24$, which is Fact 3.10 and Remark 4.2; the two trace formulas of Fact 3.12 for $2 \leq D \leq 24$; the fill and the three residue-class sums of Lemma 3.6 for $2 \leq D \leq 12$; Proposition 3.7 for $3 \leq D \leq 7$ at every offset $1 \leq m \leq r_D$ over windows through $L = 2n_D + 3$, together with the sharpness witness of Remark 3.8 at $D = 4$, $m = 2$; the anchors of Fact 3.13 at $D = 2, 3, 4, 5, 6$, including the matrix, its characteristic polynomial, the census, and $\log_3((9 + \sqrt{33})/2)$ to six places; the spectral ratios of Conjecture 3.15 for $4 \leq D \leq 20$, whose monotonicity and agreement with $(D+2)/(D-2)$ are asserted over $6 \leq D \leq 20$; and both sign checks of Conjecture 3.14, floating-point roots for $2 \leq D \leq 20$ and the exact rational sign of $\chi_D(f_D/3)$ for $2 \leq D \leq 30$. Characteristic polynomials are computed by the Faddeev–LeVerrier recursion over exact rationals; only the two spectral checks and the $D = 3$ dimension use floating point, and the spectral checks do so on integer polynomials rescaled by $f_D/3$ so that all roots have modulus near one.

`scripts/figure.py` runs instantly and redraws `figures/order.svg`, the README’s copy of Fig. 1, from the same two formulas $2D+1$ and $\lceil D/2 \rceil$.

The $D = 3$ census is OEIS [A299916](#) [8]. The ladders at $D = 4, 5, 6$ are not in the OEIS to the author's knowledge; no novelty search was run, and they are offered here only as computed values.

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